

10.5.1

EE25BTECH11033 - Kavin

Question:

Let ABC be a right triangle in which $AB = 6\text{cm}$, $BC = 8\text{cm}$ and $\angle B = 90^\circ$. BD is the perpendicular from B on AC . The circle through B, C, D is drawn. Construct the tangents from A to this circle.

Solution:

Let,

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \quad (1)$$

Let the equation of the line passing through A and C be,

$$\mathbf{n}^\top \mathbf{x} = c \quad (2)$$

The direction vector of line AC is given by,

$$\mathbf{m} = \mathbf{A} - \mathbf{C} \equiv \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (3)$$

Therefore, the normal vector of the desired line is given by,

$$\mathbf{n} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad (4)$$

$$c = \mathbf{n}^\top \mathbf{A} = \begin{pmatrix} 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \end{pmatrix} = 24 \quad (5)$$

Let D be the foot of the perpendicular from B to the line

$$\begin{pmatrix} \mathbf{m} & \mathbf{n} \end{pmatrix}^\top \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ c \end{pmatrix} \quad (6)$$

$$\begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ 24 \end{pmatrix} \quad (7)$$

$$\mathbf{m}^\top \mathbf{B} = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \quad (8)$$

$$\Rightarrow \begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} 0 \\ 24 \end{pmatrix} \quad (9)$$

The augmented matrix is given by,

$$\left(\begin{array}{cc|c} -4 & 3 & 0 \\ 3 & 3 & 24 \end{array} \right) \quad (10)$$

$$R_1 \rightarrow \frac{-1}{4}R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 3 & 3 & 24 \end{array} \right) \quad (11)$$

$$R_2 \rightarrow R_2 - 3R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 25/4 & 24 \end{array} \right) \quad (12)$$

$$R_2 \rightarrow \frac{4}{25}R_2 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 1 & 96/25 \end{array} \right) \quad (13)$$

$$R_1 \rightarrow R_1 + \frac{3}{4}R_2 \Rightarrow \left(\begin{array}{cc|c} 1 & 0 & 72/25 \\ 0 & 1 & 96/25 \end{array} \right) \quad (14)$$

$$\mathbf{D} = \begin{pmatrix} 72/25 \\ 96/25 \end{pmatrix} \quad (15)$$

To find the circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ the parameters are given by

$$\begin{pmatrix} 2\mathbf{B} & 2\mathbf{C} & 2\mathbf{D} \\ 1 & 1 & 1 \end{pmatrix}^T \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = - \begin{pmatrix} \|\mathbf{B}\|^2 \\ \|\mathbf{C}\|^2 \\ \|\mathbf{D}\|^2 \end{pmatrix} \quad (16)$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 16 & 0 & 1 \\ 144/25 & 192/25 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ -64 \\ -576/25 \end{pmatrix} \quad (17)$$

The augmented matrix is given by,

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 16 & 0 & 1 & -64 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (18)$$

$$R_1 \leftrightarrow R_2 \Rightarrow \left(\begin{array}{ccc|c} 16 & 0 & 1 & -64 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (19)$$

$$R_1 \rightarrow \frac{1}{16}R_1 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (20)$$

$$R_3 \rightarrow R_3 - \frac{144}{25}R_1 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 192/25 & 16/25 & -504/25 \end{array} \right) \quad (21)$$

$$R_2 \leftrightarrow R_3 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 192/25 & 16/25 & -504/25 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (22)$$

$$R_2 \rightarrow \frac{25}{192}R_2 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 1 & 1/12 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (23)$$

$$R_2 \rightarrow R_2 - \frac{1}{12}R_3, R_1 \rightarrow R_1 - \frac{1}{16}R_3 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (24)$$

$$\Rightarrow \mathbf{u} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} \text{ and } f = 0 \quad (25)$$

Let $\mathbf{n}^\top \mathbf{x} = c$ be the equation of tangent to the circle.
Also,

$$\mathbf{n}^\top \mathbf{A} = c \quad (26)$$

and,

$$\frac{|\mathbf{n}^\top (-\mathbf{u}) - c|}{\|\mathbf{n}\|} = r \quad (27)$$

where

$$r = \sqrt{\|\mathbf{u}\|^2 - f}$$

$$\Rightarrow \mathbf{n}^\top \mathbf{u} + c = \pm \|\mathbf{n}\| \sqrt{\|\mathbf{u}\|^2 - f} \quad (28)$$

$$\Rightarrow \mathbf{n}^\top = (\mathbf{u} + \mathbf{A}) \pm \|\mathbf{n}\| \sqrt{\|\mathbf{u}\|^2 - f} \quad (29)$$

substituten $\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}$ where m is the slope of the line.

$$\Rightarrow \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} -4 \\ 6 \end{pmatrix} = \pm 4 \sqrt{1 + m^2} \quad (30)$$

$$m = -5/12 \implies \mathbf{n} = \begin{pmatrix} 5 \\ 12 \end{pmatrix} \quad (31)$$

$$c = \mathbf{n}^\top \mathbf{A} = 72 \quad (32)$$

Therefore, the line $\begin{pmatrix} 5 & 12 \end{pmatrix} \mathbf{x} = 72$ and the y-axis are the two tangents.

